

Unit

3

Lesson

1



Beyond Imagination.

The Secret in Design.

Time Frame: [80-90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Identify the key parts of an app interface, such as labels, buttons, images, and text-to-speech blocks.
- Use MIT App Inventor to design a simple and user-friendly interface for a translation app.
- Convert a Canva design into an interactive MIT App Inventor layout.
- Reflect on how thoughtful interface design can support users with communication challenges.

Challenge questions of the lesson.

- What are the main components of an app interface that help users navigate and interact, such as labels, buttons, images, and text-to-speech features?
- How can you design a simple and user-friendly interface in MIT App Inventor that helps users easily translate messages?
- What steps do you take to convert a Canva design into an interactive layout using MIT App Inventor?
- Why is it important to think about the needs of users with communication challenges when designing an app interface? How can your design choices support these users?

SEL Relation

- **Self-Awareness:** Students reflect on their own design choices and how those choices impact the user experience, developing awareness of their creative process and decision-making.
- **Self-Management:** The lesson encourages focus, patience, and perseverance as students design, revise, and implement their app interfaces.
- **Social Awareness:** Through considering users with communication challenges, students develop empathy and understanding of diverse needs.
- **Relationship Skills:** Peer collaboration and feedback sessions build communication, active listening, and constructive criticism skills.
- **Responsible Decision-Making:** Students make thoughtful design decisions that prioritize accessibility and usability, considering the ethical impact of technology on users.

Engage



Teacher's Role:

- Gather students and create a welcoming, curious atmosphere to introduce the story.
- Read the story aloud with expression, emphasizing key ideas about app design and user experience.
- Pause occasionally to highlight important points (e.g., simplicity, icons, accessibility).
- Prompt students to think about challenges people with communication difficulties might face when using technology.
- Introduce the thinking question and encourage students to reflect and share ideas.
- Scaffold discussion by providing examples or rephrasing student answers to deepen understanding.

Expected from students:

- Listen actively to the story about Adam and Laila designing their app.
- Reflect on the importance of app appearance and functionality in helping users.
- Think critically about how design choices can impact ease of use for people with communication challenges.
- Participate in class discussion and write thoughtful responses to the thinking question.

Parts of the book included.



Let's Think!



How can the way an app looks and works make it easier—or harder—for someone with communication challenges to use it?

Story



Objective:

- Begin by gathering the class and setting a collaborative and curious tone.
- Say warmly with interest:
 - “Today, Adam and Laila are designing an app to work with their smart glove. Let’s listen to how they plan to make it simple and easy to use for everyone.”
- Read the story aloud with clear expression and a moderate pace to engage students.
- Use different tones for Adam and Laila to bring the dialogue to life and maintain attention.
- Pause after key sentences to ask quick reflective questions like:
 - “Why do you think buttons and labels are important for an app?”
 - “How might icons like speakers or flags help users?”
- Encourage students to imagine themselves designing an app that anyone—like Samir—can use easily and confidently.
- Prompt them to think about how design can help or hinder communication.

Possible Strategies for Implementation:

1. Activate Prior Knowledge:

- Ask students if they have ever used an app that was hard or easy to navigate and why.

2. Model Expressive Reading:

- Demonstrate how to read the dialogue with character voices and emotion to engage students.

3. Guided Discussion:

- After reading, facilitate a brief discussion on the purpose of app design elements (buttons, colors, icons).

4. Use Visual Aids:

- Show images of app icons, buttons, or simple UI designs to support understanding.

5. Relate to Real-Life Use:

- Encourage students to think about who might benefit from accessible design, fostering empathy.

Let's Think!



Objective:

Students will analyze how app design elements affect usability, particularly for people with communication challenges. They will reflect on ways to improve accessibility through thoughtful visual and interactive features.

Answers:

1. How can the way an app looks and works make it easier for someone with communication challenges to use it?

- The app should have clear buttons that are easy to press, simple and consistent colors that don't confuse the user, and labels that explain what each button does. Icons like speakers or flags help quickly identify functions, making the app intuitive. The design should avoid clutter and be easy to navigate so users feel confident.

2. How can the app design make it harder for someone to use it?

- If the app has too many buttons, confusing colors, or no labels, it can be overwhelming. Small buttons or unclear icons can make navigation difficult. Complex menus or unclear language can confuse users, especially those with communication difficulties.

Possible Strategies for Implementation:

1. Set Clear Purpose:

- Explain to students that they will think about how design impacts ease of use for people who face communication challenges.

2. Think-Pair-Share:

- Have students write their initial thoughts individually in the space provided.
- Then pair up to discuss their answers with a partner, encouraging sharing of ideas.

3. Prompt with Examples:

- Offer examples such as:
 - "Using clear buttons and simple colors can make navigation easier."
 - "Labels help users understand what each button does without confusion."
 - "Icons like speakers or flags can quickly show functions like sound or language."

4. Facilitate Whole-Class Discussion:

- Invite pairs to share highlights from their discussion and guide conversation toward key ideas about accessibility and simplicity.

5. Encourage Empathy:

- Ask students to put themselves in the shoes of someone who might find apps difficult and consider what design changes would help them.

Explorer's Toolkit



Teacher's Role:

- Introduce the activity by explaining that students will design their own app screens using an online drawing tool.
- Guide students to consider important app elements like buttons, labels, colors, and icons that help users navigate easily.
- Provide a brief demo or walkthrough of the Excalidraw website interface to familiarize students with the tools.
- Encourage creativity but remind students to focus on simplicity and accessibility for users with communication challenges.
- Monitor progress, offering help with the drawing tool and prompting deeper thinking about user experience.
- Facilitate sharing and peer feedback once designs are drafted.

Expected from students:

- Access and navigate the Excalidraw website to plan and draw an app screen layout.
- Thoughtfully decide on placement and design of buttons, labels, and images to support easy use.
- Use the drawing tools to create clear and organized app designs.
- Participate in peer discussions or sharing sessions to explain design choices and receive constructive feedback.
- Reflect on how their design helps users with communication challenges.

Parts of the book included.

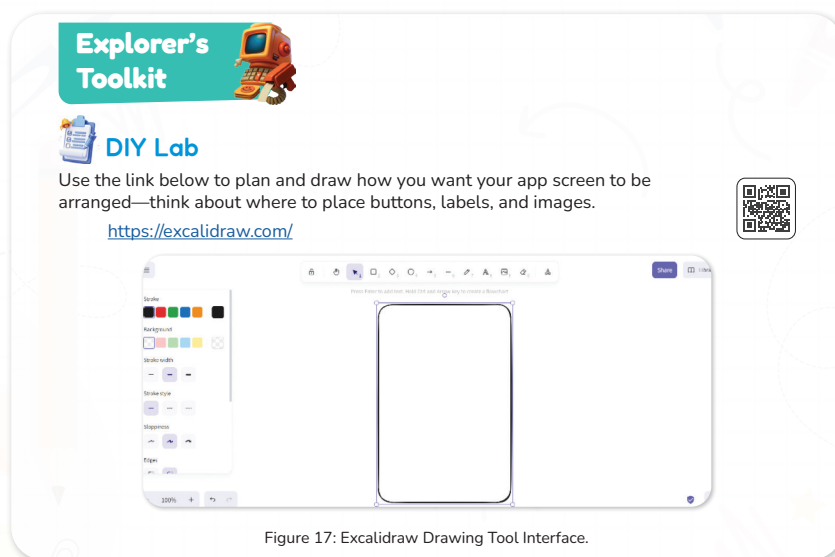


Figure 17: Excalidraw Drawing Tool Interface.



DIY Lab

Objective:

Students will apply design principles to plan an app interface that is user-friendly and accessible. They will practice visual communication and layout skills by creating a digital mockup emphasizing clear buttons, labels, and icons.



Site Link: <https://excalidraw.com/>

Activity instructions:

1. Step 1: Introduce the Task.

- Explain:
 - “Imagine someone using your smart glove needs a mobile app to see messages or send alerts.”
 - “You’ll now sketch what the app screen would look like.”

2. Step 2: Open Excalidraw.

- Direct students to: <https://excalidraw.com>
- Encourage them to draw a phone or tablet screen outline
- Use shapes/text to add:
 - Buttons (e.g., “Speak”, “Clear”, “Alert”).
 - Labels or text areas (to show received messages).
 - Icons or sensor status indicators.



3. Step 3: Design Guidance.

- Encourage students to consider:
 - Where is the most important message shown?
 - Is the screen easy to understand for someone with hearing or speech difficulties?
 - Are there visual cues (colors, symbols, etc.)?

Possible Strategies for Implementation:

1. Demonstrate Tool Usage:

- Walk through the basics of Excalidraw — drawing shapes, adding text, changing colors, and inserting icons.

2. Provide Design Guidelines:

- Remind students to prioritize simplicity, clear labeling, and meaningful icons relevant to app functionality.

3. Scaffold with Examples:

- Show sample app screen sketches or templates to guide students’ planning.

4. Encourage Iterative Design:

- Suggest students start with a rough sketch, then refine details based on peer or teacher feedback.

5. Promote Accessibility Thinking:

- Have students consider how their design choices affect users who might struggle with communication or technology.

6. Facilitate Reflection:

- End with a class discussion or journal prompt on what makes an app easy or difficult to use.

Explain



Teacher's Role:

- Introduce the video by explaining that students will learn how to upgrade a basic user interface into a professional design.
- Set the purpose for watching: encourage students to observe specific design improvements like layout, color harmony, and icon usage.
- Pause the video at key moments (if possible) to highlight important points and check for understanding.
- Encourage note-taking or sketching ideas during the video for later application.
- Facilitate a brief discussion after watching to reinforce key takeaways about professional UI design principles.
- Support students in connecting these design ideas to their own app screen projects.

Expected from students:

- Watch the video attentively, focusing on design elements that improve user experience.
- Take notes or sketch design ideas inspired by the video.
- Participate in discussion or reflection about how they can apply these improvements to their own work.
- Ask questions or seek clarification on any design concepts that are unclear.

Parts of the book included.



Screen Lessons

Click the link to discover how to upgrade your basic user interface into a professional-looking design.

<https://drive.google.com/file/d/1uxadDhN0M4xbnnwvu6z5TXK3Q-ylp-BL/view>



Screen Lessons

Objective:

Students will understand the key principles of professional user interface (UI) design and how to enhance a basic app screen to improve usability and visual appeal. They will identify design elements such as layout, color schemes, icons, and labels that contribute to an effective interface.

Video Link: https://drive.google.com/file/d/1uxadDhN0M4xbnnwvu6z5TXK3Q-yIp-BL/view?usp=drive_link.



Possible Strategies for Implementation:

1. Set a Viewing Purpose:

- Before playing, tell students to focus on how the video transforms a simple UI into a polished one.

2. Active Watching:

- Encourage students to jot down or sketch any design ideas or questions that arise.

3. Segmented Viewing:

- Pause the video at critical design changes to discuss why those changes improve usability or aesthetics.

4. Connect to Prior Knowledge:

- Ask students to relate the new design concepts to their previous app design work.

5. Post-Viewing Reflection:

- Facilitate a class conversation about which improvements they find most useful or surprising.

6. Application Prompt:

- Prompt students to think about how they will incorporate these design elements into their own app mockups.

Elaborate



Teacher's Role:

- Provide clear instructions and demonstrate how to use Canva to design app interfaces, emphasizing saving parts as separate PNG files.
- Monitor students' progress as they create their own app interface designs, providing support and encouragement.
- Facilitate the quiz activity by reviewing questions, ensuring students understand each item and encouraging critical thinking.
- Guide students during the Showcase activity to import their designed images into MIT

App Inventor and build the user interface.

- Support students in reflecting on accessibility features they added and how those improve usability for people with communication challenges.
- Encourage peer collaboration and constructive feedback during the showcase.
- Scaffold for students needing technical or design help, ensuring all can participate meaningfully.

Expected from students:

- Actively engage with the Canva design platform, exploring creative ways to arrange buttons, labels, and icons for clarity and accessibility.
- Complete the quiz with honest effort, thinking critically about user interface design best practices.
- Import and organize their design components within MIT App Inventor as directed.
- Reflect thoughtfully on their design choices and articulate how their app design supports users with communication needs.
- Collaborate respectfully with peers, giving and receiving feedback during the showcase.
- Demonstrate problem-solving and perseverance during the design and building process.

Parts of the book included.



Assessment



Apprentice



1. What is the recommended format for exporting individual UI components from Canva to use in MIT App Inventor?
 - A. .DOCX.
 - B. .SVG.
 - C. .PNG with transparent background.
 - D. .GIF.
2. In MIT App Inventor, where do you upload the exported images from Canva?
 - A. In the Blocks section.
 - B. In the Media section of the Designer tab.
 - C. In the Assets Manager.
 - D. In the Emulator tab.
3. Why is it not ideal to use a full-screen design as a single background image in MIT App Inventor?
 - A. It takes too long to upload.
 - B. It increases the app size.
 - C. It doesn't allow interactive elements like buttons to work.
 - D. It changes the screen color.

Builder



Click the Canva link to start designing your own interface. Be sure to save each part (like buttons and icons) as a separate PNG file.

<https://www.canva.com/>



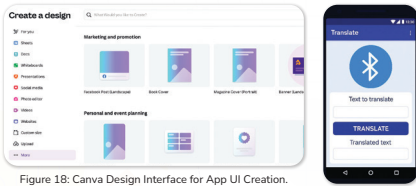


Figure 18: Canva Design Interface for App UI Creation.





Apprentice

Objective:

Students will demonstrate understanding of how to export individual UI components from Canva for use in MIT App Inventor. They will identify where to upload images within the App Inventor interface and understand best practices for designing app backgrounds to optimize performance and usability.

Answers:

1. What is the recommended format for exporting individual UI components from Canva to use in MIT App Inventor?

- Correct Answer: C) .PNG with transparent background
- Explanation: PNG format supports transparency, which is important for layering UI components without blocking other elements.

2. In MIT App Inventor, where do you upload the exported images from Canva?

- Correct Answer: B) In the Media section of the Designer tab
- Explanation: The Media section is where images and sounds are managed for use in the app design.

3. Why is it not ideal to use a full-screen design as a single background image in MIT App Inventor?

- Correct Answer: C) It doesn't allow interactive elements like buttons to work
- Explanation: A full-screen background image may block interactive elements, preventing users from pressing buttons or controls properly.

Possible Strategies for Implementation:

- **Pre-Assessment Review:** Briefly review with students the steps of exporting images from Canva, emphasizing the importance of file format (.PNG with transparent background).
- **Multiple Choice Quiz:** Present the quiz questions one by one, allowing students time to think or discuss in pairs before answering.
- **Clarify Concepts:** For each question, explain why certain answers are correct or incorrect to reinforce learning.
- **Encourage Reflection:** After the quiz, prompt students to share why exporting in the right format matters and how interface design affects app functionality.
- **Use Real Examples:** Show screenshots or demo how to upload images in the Media section of App Inventor and why full-screen background images might hinder interactivity.
- **Follow-up Discussion:** Discuss how attention to detail in UI design impacts user experience, especially for users with accessibility needs.



Objective:

Students will design a user-friendly interface for their translation app using Canva. They will learn how to break down a full layout into individual components (e.g., buttons, icons, backgrounds) that can be reused in MIT App Inventor.

Activity Instructions:

1. Start Canva

- Open www.canva.com
- Select a custom size layout (e.g., 480x800 px) to match a phone screen.



2. Design the Layout

- Add elements: text labels, buttons, icons, title area.
- Arrange them using Canva's alignment tools (use vertical and horizontal guides for structure).
- Use simple shapes and bold icons to support readability.

3. Export Components

- Click on each item (e.g., button, icon).
- Use "Download" → PNG → Transparent background → "Download Selected".
- Save each component separately (e.g., button_start.png, icon_translate.png).

4. Organize Files

- Save all files in a folder with clear names.
- Check that no element has a background unless intended.

Possible Enhancements:

- Add alt text to each element to discuss its purpose.
- Use peer review: Have students exchange designs and provide feedback on usability.
- Include a reflection prompt:
 - "What did you change in your design to make it clearer or easier to use?"

Possible Strategies for Implementation:

1. Model First – Think Aloud Design.

- Begin by projecting Canva on the board and designing a simple interface live. Narrate your thinking:
 - "I'm adding a big 'Speak' button in the middle so it's easy to find."
 - "I'm using an icon instead of a word here because not everyone reads easily."
- This helps students see how function and accessibility guide design choices.

2. Chunk the Task – Design One Element at a Time.

- Guide students to design one component at a time (e.g., just a button). After they complete and export one item, they can move to the next.
- This avoids overwhelming students with a full screen layout all at once.

3. Use a Design Checklist.

- Provide a printed or digital checklist:
 - Title area
 - Speak button
 - Translate icon
 - Clear background
 - Saved as PNG
- This ensures students stay focused and follow export requirements.

4. Peer Feedback – 2 Stars and a Wish.

- After students design and export their elements, pair them up to review each other's UI. Ask them to give:
 - Two things they like.
 - One suggestion (wish) to improve clarity or accessibility.

5. Connect to Real Users – Empathy Lens.

- Ask students to imagine they're designing for a user who:
 - Can't read
 - Doesn't speak the same language
 - Is using the app in a hurry
- Then ask: "Does your interface help them succeed? What can you improve?"

6. Accessibility Icons Hunt (Optional Warm-Up).

- Before the design activity, challenge students to explore accessibility icons (e.g., hearing aid icon, speaker icon, microphone icon). Let them use these as part of their app.

7. Gallery Walk

- Display the exported component images or final layouts and let students "walk" around the room or screen to give sticky note feedback or emojis digitally. Helps reinforce design thinking and peer learning.

Evaluate



Teacher's Role:

- **Design Mentor**
 - Guide students in translating their UI sketches into a working design in MIT App Inventor.
 - Reinforce design principles like clarity, accessibility, and layout balance.
- **Accessibility Advocate**
 - Prompt students to consider how their interface can be used by someone who cannot read or speak easily.
 - Encourage use of features like icons, voice buttons, or simplified language.
- **Technical Guide**
 - Support students in navigating MIT App Inventor's components (e.g., buttons, labels, images).
 - Troubleshoot basic UI building issues and remind them how to preview their design.
- **Facilitator of Reflection**
 - Ask reflective prompts like:
 1. "What challenge did you consider in your design?"
 2. "How does your UI help all users, regardless of ability?"

Expected from Students:

- **Build Their App Interface**
 - Open MIT App Inventor and create their app screen layout.
 - Use appropriate UI components (e.g., images, buttons, labels) as per their previous sketches.
- **Apply Accessibility Features**
 - Incorporate at least one feature that supports non-verbal or low-literacy users (e.g., symbols, colors, or audio cues).
 - Ensure the UI is simple and clear for all users.
- **Reflect and Justify Design Choices**
 - Respond to the reflection prompt in complete sentences, explaining what feature they included for accessibility and why.



Showcase

Objective:

Students will import and arrange their Canva-exported interface components (buttons, labels, icons) into MIT App Inventor, and reflect on how their design supports users with different communication needs (e.g., non-readers, non-speakers).

Mobile App Resources: https://drive.google.com/file/d/1_4S66hlumShkKyMIKIwUCKxeqr59pZT/view?usp=drive_link



Activity Instructions:

1. Open MIT App Inventor:

- Go to MIT App Inventor and log in.
- Create a new project and name it (e.g., GloveTranslatorUI).

2. Upload Design Assets:

- In the Designer tab, go to the Media section.
- Click "Upload File" and add each PNG image you exported from Canva (buttons, icons, backgrounds).

3. Build the Interface:

- Drag and drop Image components to match your Canva layout.
- Use HorizontalArrangement and VerticalArrangement for organizing.
- Add Label and Text-to-Speech components if needed.
- Optional: Add Buttons with transparent text overlays to allow interaction.

4. Match Layout to Design:

- Align and resize each element to fit your original design concept.
- Adjust screen color, text font, or spacing for better visual clarity.

5. Reflection Question (Class Discussion or Writing Prompt):

- "What feature did you add in your design to support users who may not be able to read or speak easily?"

Possible Strategies for Implementation:

1. Design Walkthrough Before Building:

- Start by projecting a sample Canva design and showing how each PNG is placed inside MIT App Inventor. Highlight the use of layout components (horizontal/vertical arrangements) to scaffold screen design.

2. Student Pair Review:

- Let students work in pairs to preview each other's designs before uploading. They can give feedback on accessibility (e.g., "Can someone with vision or speech challenges use this easily?").

3. Model Reflection:

- Use think-aloud modeling to show how to answer the reflection question. For example:
- “I added icons and a text-to-speech button, so someone who can’t read can still understand what’s being said.”

4. Accessibility Checklist:

- Provide a simple checklist students can use as they design (e.g., “Do I have clear icons? Do buttons make sound or show text?”).

Rubric: Building an Accessible App Interface:

Criteria	Excellent (4 Point)	Good (3 Points)	Needs Improvement (2 Points)	Beginning (1 Point)
Design Accuracy	All components are clearly arranged and match the Canva design perfectly.	Most components are arranged correctly and resemble the Canva layout.	Some components are misplaced or missing from the layout.	Layout is incomplete or does not match the design plan.
Image Integration	All PNGs are properly uploaded, sized, and used purposefully in the UI.	Most PNGs are used well, with a few minor size or placement issues.	Some PNGs are missing or incorrectly used.	Few or no images are integrated correctly.
Accessibility Features	Design includes excellent features to support users with communication needs.	Design includes some accessibility elements (e.g., icons, speech, colors).	Design includes minimal accessibility support.	No consideration of accessibility is shown.
Layout & Clarity	Screen is organized, easy to understand, and visually appealing.	Screen is mostly organized and clear, with minor improvements needed.	Layout is confusing or cluttered.	Layout is disorganized and hard to navigate.

Reflection on Design	Clearly explains how the design supports diverse users (written or oral).	Reflects on design with some relevant points.	Reflection is incomplete or lacks depth.	No reflection provided.
Effort & Creativity	Shows exceptional creativity and care in the app design process.	Shows thoughtful design and good effort.	Shows some effort, but lacks originality or attention to detail.	Minimal effort; rushed or incomplete work.

It's time to move to the improve part and add this.



Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. Introduce the Reflection

- Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."

2. Guide Students Through the Checklist.

- Read each statement aloud one at a time.
- After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
- Example prompts:
 - “Did you learn how people share ideas without talking?”
 - “Do you know how tools can help people talk?”

3. Student Completion

- Let students check off or color the “Now I Can” boxes they feel good about.
- Remind them: “It’s okay if you didn’t check all of them. We’re all still learning.”

4. Optional Extension

- Ask students to circle the one they’re most proud of.
- Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. Next Generation Science Standards (NGSS):

- **MS-ETS1-1:** Define problems that can be solved through engineering design.
- **MS-ETS1-2:** Evaluate competing design solutions based on criteria and constraints.

2. Common Core State Standards (CCSS) – English Language Arts:

- **CCSS.ELA-LITERACY.SL.6.1:** Engage effectively in collaborative discussions with diverse partners on grade 6 topics and texts.
- **CCSS.ELA-LITERACY.W.6.2:** Write informative/explanatory texts to examine a topic and convey ideas clearly.

3. International Society for Technology in Education (ISTE):

- **Empowered Learner:** Use technology to set goals, work toward achieving them, and demonstrate learning.
- **Computational Thinker:** Develop and employ strategies for understanding and solving problems in ways that leverage technology.